



GRADE 10 · MODULE 1.2

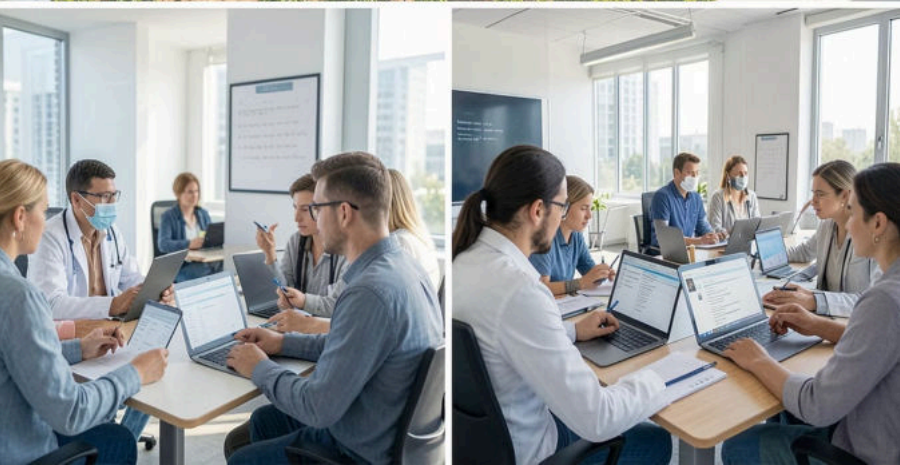
Applications of ICT in Daily Life

Grade 10 - ICT - Module 1.2

"Today, you could hardly come across any person or a place that does not use ICT"

Learning Objectives

1. Identify ICT applications in education
2. Understand ICT in health sector
3. Explain ICT in agriculture and farming
4. Describe ICT in business and manufacturing
5. Recognize ICT in transport and entertainment



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ICT in Education - Overview

Days are gone when we were stuck among piles of books to collect required information. Today you have the luxury of seeing the entire world at your convenience due to ICT advancements.

In the Classroom

Education - Anywhere Anytime

Teaching Aid for Teachers

**Learning Management Systems
(LMS)**

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ICT in Education - In the Classroom

Ways ICT is used in the classroom:

- Presentations
- Videos on experiments
- Creation of images and video
- Desktop publishing of magazines, letters, and documents
- Educational games (Edutainment)
- CD-ROM media for learning
- Gathering educational information on the internet

? Interactive Element:

How many of these have you experienced in your classroom?

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ICT in Education - Anywhere Anytime Learning

If you have a computer with internet connection, you can study while at home or any convenient place.

Educational Websites

Website	Purpose
www.schoolnet.lk	School network resources
www.nenasala.lk	Nenasala knowledge centers
www.e-thaksalawa.moe.gov.lk	Government e-learning platform
www.edulanka.lk/Web-Patashala	Online learning
www.vidumanpetha.com	Educational resources

Web Based Training (WBT)

Contact with a teacher/trainer via internet

Advantages:

- Minimize travel difficulties
- Save time and money

Activity:

1. Visit the websites listed above
2. List areas that support your knowledge
3. Compare with traditional teaching methods
4. Visit BBC Bitesize (www.bbc.co.uk/education)



ICT in Education - Teaching Aid & LMS

Teaching Aid for Teachers

- Using pictures, animations, and audio-visuals to explain difficult concepts
- Making lessons interesting with presentations
- Organizing lessons using computer
- Obtaining subject-relevant information

Learning Management System (LMS)

Definition: "System used to manage school system or higher education system"

- Need internet connection to access LMS
- Must register on official website

LMS Services Table:

For Students	For Management
Use learning units anytime/anywhere	Add quality learning materials
Upload assignments from home	Supervise activities and publish results
Submit queries through forums	Maintain updated student/teacher information
Participate in co-curricular activities via video	Send/receive emails to parents and officials
Parents can monitor progress from home	Automatic ringing systems (bells)



ICT in Education - Distance Learning

With ICT development, today a person from any country can connect with a university and pursue higher education at low cost.

Features of Distance Learning:

- No daily timetables - learn at your convenience
- Digital library facility on joining a course
- Online assignments and quizzes
- Contact with teacher online
- Easy to obtain teacher consultations

? Discussion Point:

How does distance learning help students in remote areas?



ICT in Health Sector - Diagnostic Equipment

Many difficult processes in the medical field have become efficient through modern equipment.

Machine	Purpose
CAT Scan (Computerized Axial Tomography)	Creates 3D images of body parts for diagnosis
MRI Machine (Magnetic Resonance Imaging)	Creates digital images of internal organs using magnetic fields and radio waves
ECG Machine (Electrocardiogram)	Monitors heart beat by recording electrical impulses
Cardiac Screening	Displays heart physiology and movements, detects vein issues
EEG Machine (Electro-encephalography)	Records brain activities using electrical probes
Blood Sugar Tester	Analyzes blood sample to determine glucose level
Blood Pressure Monitor	Measures BP at rest and during activity



ICT in Health - Telemedicine

Definition:

"The use of ICT to examine and provide healthcare to a patient who is far away from the hospital is called Telemedicine."

Five Types of Telemedicine:

Type	Description
Emergency Telemedicine	Emergency care by trained person consulting specialist via network (remote village, plane, ship)
Home Health Medicine	Monitoring patient at home using networked system
Telemedicine Consultation	Specialist consultation at ETU when not physically present
Telesurgery (Remote Surgery)	Doctor performs surgery consulting specialist in different location
Medical Teletraining	Training medical professionals remotely using telecommunication

Advantages:

- Patient can stay home
- Reduces travel
- Specialist access from anywhere

ICT in Agriculture - Farming

ICT has created a new revolution in agriculture, animal husbandry, and fisheries industries.

Agricultural Machines and Technologies:

Meteorological Devices Assess weather, climate, rainfall, wind direction	Automated Insect Control Monitor insect population, growth, movement	Field Condition Devices Measure soil fertility and humidity
Drip Irrigation Control water supply automatically	Automatic Weed Remover Identifies crops and weeds separately	Robotics Seedlings Planter Plants seedlings in orderly manner
Crop Harvesting Robots Monitor growth and harvest	Greenhouses Control light, moisture, air; protect from natural disasters	





ICT in Farmhouse Management

RFID (Radio Frequency Identification Device)

Help in identifying and counting animals and locating them in large areas

Automated Milking and Examination

Monitor health status of cows and quality of milk

ICT in Fishing Industry

- Sensors placed in sea convey information on fish concentration
- Connected to computers in fishing trawlers via internet

Additional Technologies

- Closed circuit TV for farm monitoring
- Computers in managing farm operations

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ICT in Manufacturing Industry

ICT is used to manufacture high quality products and reduce production cost by minimizing human labour.

Past vs Present Comparison:

Past	Present
High level of human labour needed	Most work performed by robots
Time-consuming processes	24-hour service
Human errors possible	High efficiency and accuracy
Hygiene concerns	Better hygiene

Benefits of Robots:

24-hour service

Never get tired

Efficiency

Accuracy

Hygiene





ICT in Business

Video Conferencing

Different business communities who are geographically separated can convene face-to-face meetings while at their own premises

Advantages:

- No need to convene at special venue
- Saves time
- Reduces travel expenses

Human Resource Management

- Fingerprint scanner for attendance registration
- Card Reader for Identity Management
- Records: moving in/out times, salary preparation, leave records

e-Banking System

- ATM (Automatic Teller Machines) - withdraw cash anytime anywhere
- Inter-banking transactions within Sri Lanka or abroad
- Mobile banking - pay utility bills, check account balance



ICT in Transport

Closed Circuit TV (CCTV):

Monitor road traffic, accidents, and illegal activities

Traffic Light Control System:

Automated lights positioned at road junctions

Minimize accidents by controlling vehicle traffic and pedestrians

Parking Identification Placard:

- Entrance pass stuck on windshield
- Sensors read placard and open gates for pre-registered vehicles
- Prevents traffic build-up at gates

? Activity Box:

Observe other instances where ICT is used in business and manufacturing and compile a list



ICT in Entertainment

Devices and Equipment Provide:

- Listen to music to overcome monotonous life
- Watch missed TV programmes or movies
- Exchange information through social media
- Watch high definition video
- Surf the internet
- Educational games for children
- Read e-books on internet
- Store favourite photographs

Key Point:

"Various devices and equipment have been invented to provide some peace of mind to human beings spending a busy life today."

Module 1.2 Summary - ICT Applications in Daily Life

ICT Applications Across Sectors:

Sector	Examples
Education	Classroom tech, online learning, LMS, distance education
Health	Diagnostic machines, telemedicine, remote monitoring
Agriculture	Smart farming, automated irrigation, greenhouse tech
Business	Video conferencing, HR systems, e-banking
Transport	CCTV, traffic control, automated parking
Entertainment	Music, movies, social media, e-books

Review Questions:

1. Name 3 ways ICT is used in the classroom
2. What is telemedicine?
3. Name 3 agricultural machines that use ICT
4. What are the benefits of using robots in manufacturing?
5. How does ICT help in banking?

Activity:

"Observe ICT applications in your daily life and make a list"